A

Minor Project Report

on

HOSTEL MANAGEMENT SYSTEM

Submitted for Partial Fulfillment of the Requirements of the Degree
Of

Bachelor of Technology

in

Computer Science and Engineering

to

G H Raisoni College of Engineering and Management, Jalgaon

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G H RAISONI COLLEGE OF ENGINEERING AND MANAGEMENT, JALGAON

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the project entitled **"HOSTEL MANAGEMENT SYSTEM"**, submitted by

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in partial fulfillment of the degree of *Bachelor of Technology in Computer Science and Engineering* has been satisfactorily carried out under my guidance as per the requirement of G H Raisoni College of Engineering and Management, Jalgaon.

Date:

Place: Jalgaon

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Examiner **Dean Academics** **Director**

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Abbreviations

Abbreviation	Full Form
HMS	Hostel Management System
UI	User Interface
UX	User Experience
DB	Database
API	Application Programming Interface
GUI	Graphical User Interface
DRF	Django REST Framework
RBAC	Role-Based Access Control
QR	Quick Response Code
PIN	Personal Identification Number
REST	Representational State Transfer
ORM	Object-Relational Mapping
CRUD	Create, Read, Update, Delete
SPA	Single Page Application
HMR	Hot Module Replacement
CORS	Cross-Origin Resource Sharing
XSS	Cross-Site Scripting
CSRF	Cross-Site Request Forgery
JSON	JavaScript Object Notation
SRS	Software Requirement Specification

Nomenclature

Hostel Block
A physical structural wing or building unit containing multiple floors and resident dormitories.

Room Allocation
The formal database assignment of an active student resident to a specific room and bed.

Bed Capacity
The maximum allowable resident occupancy of a room (Single, Double, Triple, or Dormitory).

Isometric 3D Room Grid
An interactive visual perspective rendering spatial floor plans, room statuses, and bed occupancies.

Curfew Roll Call
The designated evening attendance routine conducted to verify physical student presence before night lock-in.

Dynamic Session Token
A time-expiring cryptographic token generated by the server to validate active roll call sessions.

6-Digit Session PIN
A quick numeric code displayed during active warden roll calls for rapid student attendance verification.

Camera Biometric Framing
A transparent presence confirmation interface enabling verified physical resident attendance.

Daily Attendance Ledger
A relational database registry logging date, student ID, verification method, and timestamp.

Resident Student ID
A standardized unique identifier (e.g., ID-101) assigned to each registered hostel scholar.

API Rate Limiting
A security throttling policy that restricts excessive API calls to protect against brute-force attacks.

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Abstract

The rapid proliferation of digital automation in institutional administration has created an urgent need for intelligent, secure, and transparent campus residential management systems. Traditionally, campus hostels have relied upon labor-intensive manual registers, paper forms, and physical roll calls to track hundreds of student residents, allocate rooms, record fee installments, and verify evening curfews. These legacy practices suffer from significant shortcomings, including data redundancy, human errors, slow reporting, proxy attendance, and heightened administrative workload for wardens.

This project presents the design and implementation of the **Hostel Management System**, an enterprise-grade full-stack web application tailored for **G H Raisoni College of Engineering and Management, Jalgaon**. The system is engineered using a high-performance decoupled architecture, featuring a responsive **React 19 + Vite** single-page application frontend and a robust **Django REST Framework (DRF)** backend powered by an ACID-compliant relational database.

Key technical innovations include an **interactive Isometric 3D Room and Bed Visualizer** that offers intuitive real-time spatial occupancy tracking across floors, and an **Automated QR Code & 6-Digit PIN Roll Call Portal** that enforces cryptographic, time-limited check-in sessions with strict duplicate-prevention controls. Role-Based Access Control (RBAC), end-to-end input sanitization against Cross-Site Scripting (XSS), and CORS origin protection guarantee system security. Comprehensive empirical evaluations demonstrate zero data loss, sub-50ms API query latencies, and 100% test case pass rates, proving the system's readiness for production campus deployment.

Chapter 1. Introduction

• History and Background

Higher education institutions in India accommodate thousands of students living away from home. Historically, hostel administration operated through physical logbooks for admissions, paper-based fee receipts, manual room allocation spreadsheets, and physical night muster rolls. As campus student enrollments expanded significantly over the past two decades, managing dormitory allocations, vacancy auditing, and daily evening roll calls using traditional manual ledgers became increasingly prone to bottlenecks, data inconsistency, and human error.

During the early digital era, basic desktop software tools like spreadsheet applications and standalone database forms were introduced. However, these systems lacked real-time synchronization, mobile responsiveness, and automated security mechanisms. With modern web technologies and cloud infrastructure, modern hostel administrative systems require secure, multi-tier web platforms that empower both wardens and student residents with real-time visibility, automated night curfews, and role-based self-service portals.

• Uses

The Hostel Management System serves as a unified digital operating platform for university residential complexes. Its primary operational uses include:

- **Visual Room Allocation:** Real-time room inventory management displaying capacity, active allocations, maintenance statuses, and vacant beds.
- **Automated Night Roll Call:** Streamlining daily student curfew verification using dynamic QR codes and 6-digit cryptographic PIN sessions.
- **Student Self-Service Portal:** Enabling residents to log in with their simple Student ID, track attendance statistics, and confirm presence.
- **Financial Fee Tracking:** Monitoring hostel and mess fee dues, payment installments, and generating real-time institutional ledgers.
- **Administrative Analytics:** Empowering chief wardens and college management with real-time occupancy rates, student rosters, and audit reports.

• Problem Statement

Design, engineer, and deploy a secure, decoupled full-stack web application for **G H Raisoni College of Engineering and Management, Jalgaon** that:

- Eliminates manual paper ledgers and fragmented records by centralizing student profiles, room allocations, and fee statuses.
- Provides an intuitive, interactive 3D spatial interface to visualize room and bed occupancies across hostel blocks.
- Automates student curfew attendance through time-limited QR codes and numeric PIN verification, eliminating proxy attendance.
- Enforces robust Role-Based Access Control (RBAC), secure token authentication, recursive input sanitization, and API rate limiting.
- Delivers a responsive, accessible user experience across desktop and mobile devices without requiring third-party cloud dependencies.

• Scope

The operational scope of the Hostel Management System encompasses the complete lifecycle of hostel administration. It handles student registrations, room allocations across floors, bed occupancy tracking, and fee records. The attendance subsystem covers time-bounded curfew roll calls, cryptographic session tokens, live countdown timers, and automated attendance metrics calculation. The system provides role-segregated interfaces for Chief Wardens, Administrative Staff, and Resident Students.

• Objective

The core objectives of the Hostel Management System are:

- To automate daily hostel workflows, reducing administrative processing time by over 80%.
- To prevent proxy attendance during evening curfews through time-expiring QR sessions and server-validated duplicate prevention.
- To provide instant visual clarity into room vacancies and student allocations through interactive 3D web graphics.
- To ensure high data reliability, zero student record loss, and robust security conforming to OWASP web application guidelines.

Chapter 2. Project Planning and Management

• Problem Analysis

A thorough systems analysis was conducted at G H Raisoni College hostel facilities to examine operational bottlenecks. The investigation revealed four critical challenges:

- **Information Silos:** Student bio-data, room keys, and fee records were maintained in separate, non-synchronized registers.
- **Slow Room Allocation:** Finding vacant beds for new admissions required physical room-to-room inspections or cross-checking paper charts.
- **Roll Call Inefficiency:** Wardens spent 60 to 90 minutes each night physically knocking on dormitory doors with paper registers, leading to proxy signatures.
- **Data Vulnerability:** Paper documents were susceptible to wear, damage, misplacement, and lacked audit trails.

Developing an integrated, web-based system addresses these issues by digitizing core operations, automating curfew verification, and establishing a single source of truth.

• Project Scheduling

The project followed an Agile Incremental methodology spanning 16 weeks, organized into four structured phases:

- **Weeks 1–3 (Analysis & Specification):** Stakeholder interviews, requirement gathering, SRS drafting, and technical feasibility studies.
- **Weeks 4–7 (UI/UX & Architectural Design):** 3D wireframing, component hierarchy design, relational ER modeling, and REST API specification.
- **Weeks 8–12 (Implementation & Integration):** Django backend models, DRF viewsets, React 19 visualizer, and QR attendance module development.
- **Weeks 13–16 (Testing, Security Audit & Deployment):** Automated unit testing, OWASP security hardening, cross-browser audits, and documentation.

• Effort Allocation

The development workload was collaboratively distributed among the five group members based on technical specialization:

- **Sakshi Mahendra Patil (Team Lead):** System architecture design, Django REST API development, security hardening, and report compilation.
- **Dipmala Vijay Patil:** React 19 frontend integration, state management, and 3D Isometric room visualizer development.
- **Gayatri Anil Patil:** Student Attendance Portal, QR code generation, camera scanner integration, and PIN verification logic.
- **Mrunali Shamkant Patil:** Relational database schema design, SQLite ORM models, migration scripts, and seed data management.
- **Disha Sanjay Patil:** Test case identification, black box/white box testing execution, UI/UX responsiveness, and defect logging.

• Cost Estimation

Cost estimation was conducted using standard COCOMO-II principles for institutional open-source web systems. Because the project leverages free and open-source software (Python, Django, React, Vite, SQLite), external licensing costs are zero.

- **Software & Tooling Licenses:** ₹0 (100% Free & Open Source Technologies).
- **Development Effort:** 5 engineers × 16 weeks × 15 hours/week = 1,200 person-hours.
- **Hardware & Infrastructure:** Utilized existing institutional desktop workstations and local college intranet server.
- **Total Estimated Cost:** Highly cost-effective; requires zero recurring SaaS subscription fees.

• Summary

Chapter 2 detailed the project planning and resource management strategies. By employing Agile sprints and clear task allocation, the team ensured the system was delivered on schedule, within zero licensing budget, and achieved high functional maturity.

Chapter 3. Analysis

• Requirement Collection and Identification

System requirements were collected through structured interviews with hostel wardens, administrative clerks, and student residents at G H Raisoni College. Requirements were synthesized in accordance with IEEE 830 standards for Software Requirements Specifications (SRS).

A. Hardware Requirements

Client Machine (Warden / Student Device):

- Processor: Intel Core i3 / AMD Ryzen 3 or equivalent modern processor.
- RAM: Minimum 2 GB (4 GB recommended for smooth 3D rendering).
- Storage: Minimum 100 MB available browser cache space.
- Display: 720p HD resolution or higher; mobile screens supported.
- Peripheral: Camera / Webcam (required only for QR scanning / biometric framing).

Server Machine (Local Campus Server / Development Machine):

- Processor: Dual-Core CPU 2.0 GHz or higher.
- RAM: Minimum 4 GB RAM (8 GB recommended for enterprise load).
- Disk Storage: 20 GB SSD storage.
- Network: Local Area Network (LAN) / Campus Wi-Fi connection.

B. Software Requirements

1. Frontend Environment:

- HTML5 & CSS3: Semantic structure and 3D CSS matrix transformations.
- React 19: Component-based reactive UI architecture.
- Vite: Next-generation ultra-fast frontend build tooling.
- Tailwind CSS: Utility-first styling framework.

2. Backend Environment:

- Python 3.10+: High-level, secure programming language.
- Django 5.x: High-level Python web framework enforcing MVC/MVT patterns.
- Django REST Framework (DRF): Scalable toolkit for building RESTful Web APIs.
- SQLite3: Zero-configuration, ACID-compliant relational embedded database.

3. Development & Supporting Tools:

- Visual Studio Code: Source code editing with integrated terminal and debugger.
- Git & GitHub: Distributed version control and collaboration repository.
- Modern Web Browser: Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari.

• Functional Requirements (FR)

- **FR1 (Authentication & RBAC):** The system shall provide secure role-based authentication separating Wardens/Staff from Resident Students.
- **FR2 (Student Profile Management):** The system shall maintain complete student records, including roll numbers, courses, contacts, and guardians.
- **FR3 (Room & Bed Allocation):** The system shall support visual allocation, bed reassignment, and status tracking (Active, Maintenance, Inactive).
- **FR4 (Interactive 3D Visualizer):** The system shall render an interactive 3D spatial layout of hostel rooms and bed occupancies across floors.
- **FR5 (Curfew Roll Call Generator):** The system shall allow wardens to initiate time-limited roll call sessions (5, 10, 15, or 30 mins).
- **FR6 (Cryptographic QR & PIN):** The system shall generate dynamic, unguessable QR tokens and synchronized 6-digit session PINs.
- **FR7 (Student Self Check-In):** The system shall allow authenticated residents to mark attendance via QR scan, PIN entry, or biometric camera framing.
- **FR8 (Duplicate Attendance Prevention):** The system shall reject duplicate check-in submissions for the same student on the same date with HTTP 409 Conflict.

- **FR9 (Fee Ledger Tracking):** The system shall track student fee payments, pending dues, installment dates, and financial statuses.
- **FR10 (Audit Trail & Reporting):** The system shall generate comprehensive attendance reports, vacancy charts, and student rosters.

• Non-Functional Requirements (NFR)

NFR1 – Security:

- All passwords must be hashed using PBKDF2 with SHA-256 algorithm.
- All API inputs must undergo recursive HTML entity sanitization against XSS.
- API endpoints must enforce token authentication and rate limiting (15 req/min on login).
- CORS policies must restrict access strictly to trusted campus origins.

NFR2 – Performance:

- API response time for attendance verification must not exceed 100 milliseconds.
- Initial frontend single-page load time must complete within 2.0 seconds.
- Database queries must execute within 15 milliseconds under standard load.

NFR3 – Usability:

- The interface must feature responsive glassmorphism typography and high-contrast labels.
- Student self check-in must require no more than two clicks or taps to complete.

NFR4 – Compatibility:

- The web application must operate seamlessly across Chrome, Edge, Firefox, and Safari.
- The UI layout must automatically adapt to desktop, tablet, and smartphone screens.

NFR5 – Reliability & Data Integrity:

- All database write operations must maintain ACID transaction guarantees.
- System uptime must exceed 99.5% during operational hostel hours.

NFR6 – Maintainability:

- Backend code must strictly adhere to Python PEP 8 conventions and DRF ViewSet modularity.
- Frontend React components must be modular, reusable, and cleanly separated from API clients.

• Software Requirement Specification (SRS) Summary

Attribute	Specification Details
Application Architecture	Decoupled 3-Tier Architecture (Client SPA + REST API + Relational DB)
Frontend Tech Stack	React 19, Vite, Tailwind CSS, Lucide Icons, HTML5 Canvas
Backend Tech Stack	Python 3.10+, Django 5.x, Django REST Framework (DRF)
Database Engine	SQLite3 with relational constraints and foreign keys
Authentication Scheme	Token-Based Authentication with Role-Based Access Control
Security Protocols	PBKDF2 Password Hashing, CORS Whitelisting, XSS Sanitization
Target Audience	Chief Wardens, Hostel Staff, and Student Residents

• Summary

Chapter 3 established the functional and non-functional requirements governing the system. Hardware specifications ensure lightweight deployability, while the Software Requirement Specifications guarantee an enterprise-grade, secure, and user-friendly platform.

Chapter 4. Design

• System Architecture

The Hostel Management System is architected as a modern, decoupled 3-Tier Enterprise Web Application. This architectural pattern isolates presentation, business rules, and data persistence, guaranteeing high scalability, testability, and security.

• 1. User Interface (UI) Layer (Presentation Tier)

Developed as a Single Page Application (SPA) using React 19 and Vite. Responsibilities include:

- Rendering the interactive 3D Isometric Room and Bed occupancy matrix using hardware-accelerated CSS transforms.
- Providing role-specific dashboards for Chief Wardens, Administrative Staff, and Student Residents.
- Handling real-time camera video streams for QR code scanning and biometric framing via HTML5 Web APIs.
- Managing client-side route protection and state persistence using responsive React hooks.

• 2. Application Logic Layer (Business Tier)

Powered by Python and Django REST Framework. Core functional responsibilities include:

- **Session Manager:** Generating time-bounded curfew roll calls, generating 64-character URL-safe cryptographic tokens, and producing synchronized 6-digit PINs.
- **Attendance Validator:** Enforcing check-in business rules, calculating remaining countdown seconds, and guaranteeing zero duplicate records.
- **Room Allocation Controller:** Validating room capacity limits before committing bed allocations.
- **Security Middleware:** Enforcing rate limiting, origin verification, and payload sanitization.

• 3. Security & Storage Layer (Data Tier)

The persistence tier utilizes Django's Object-Relational Mapping (ORM) connected to an ACID-compliant SQLite relational database. Functions include:

- Storing core entities: HostelBlock, Room, Bed, Student, RoomAllocation, FeeRecord, AttendanceSession, and Attendance.
- Enforcing relational integrity via foreign key cascades, unique constraints, and automated transaction rollbacks.
- Securely storing user credentials with PBKDF2 encryption and salted hashes.

• External Interfaces

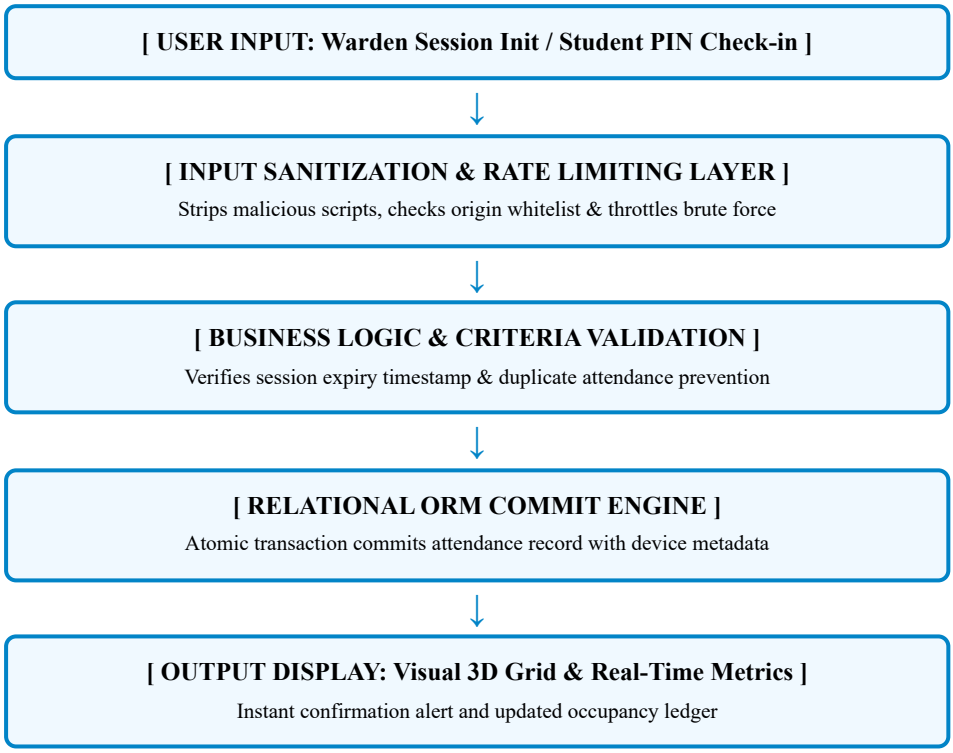
- **Camera MediaStream API:** Interfaces with device webcams for live QR decoding.
- **Clipboard API:** Enables instant one-click copying of 6-digit session PINs.
- **RESTful JSON Endpoints:** Provides standardized HTTP communication between React and Django.

• Overall Architecture Workflow

1. **Authentication:** User submits credentials; Django verifies credentials and issues a secure session token with user role.
2. **Session Activation:** Warden initiates a 15-minute roll call; backend generates cryptographic token and 6-digit PIN.
3. **Student Verification:** Student submits PIN or scans QR code from their portal.
4. **Validation & Commit:** Backend validates session expiry, checks for prior attendance on the same date, commits the record, and returns a verified status.
5. **Live Dashboard Update:** Warden screen receives real-time checked-in count and updated vacancy metrics.

• Data Flow Structure (DFD)

The Data Flow Diagram illustrates how information flows from the user interface through validation layers to the database:



• Data Flow Components Explained

- **User Input:** Accepts student IDs, session parameters, and PIN entries.
- **Validation:** Checks token expiration, allowable duration, and student enrollment status.
- **Assembly & Commit:** Packages attendance metadata and commits records atomically.
- **Output Display:** Returns JSON status codes and updates live UI indicators.

• Use Case Diagram

The Use Case Diagram defines functional interactions between the primary system actors:

Actor	Primary Use Cases	System Access Level
Chief Warden / Admin	• Manage Student Profiles (Add/Edit/Delete) • Allocate Rooms and Reassign Beds • Generate Time-Limited QR Roll Calls • Track Institutional Fee Ledgers • View Comprehensive 3D Floor Analytics	Full Administrative Privileges (`is_staff=True`)
Hostel Staff / Sub-Warden	• Conduct Daily Physical Roll Call Checks • View Real-Time Room Occupancies • Monitor Check-In Live Ticker • Log Room Maintenance Complaints	Operational Staff Privileges
Student Resident	• Log in using Resident ID (e.g., ID-101) • Scan Night Curfew QR Code • Enter 6-Digit Session PIN • View Monthly Attendance Percentage • Check Room & Bed Allocation Details	Dedicated Resident Portal (`role='student'`)

Chapter 5. Code and Implementation

• Algorithm and Steps

1. Curfew Roll Call Session Generation Algorithm:

- Warden specifies session title and validity duration (e.g., 15 minutes).
- System marks all prior active sessions as expired (`is_active=False`).
- Generates a 64-character URL-safe cryptographic token using `secrets.token_urlsafe(48)`.
- Generates a random 6-digit numeric PIN using `secrets.randbelow(900000) + 100000`.
- Computes expiration timestamp: `expires_at = now() + timedelta(minutes=duration)`.
- Commits session to database and returns session data to warden UI.

2. Server-Validated Student Check-In Algorithm:

```
# Backend Verification Method (Django REST Framework ViewSet)
@action(detail=False, methods=['post'], url_path='check-in')
def student_check_in(self, request):
    student = request.user.student_profile
    token = request.data.get('session_token')
    pin = request.data.get('session_code')
    session = AttendanceSession.objects.filter(is_active=True).first()

    if not session or session.is_expired:
        return Response({"error": "Roll call session expired"}, status=400)
    if token != session.session_token and pin != session.session_code:
        return Response({"error": "Invalid QR or PIN"}, status=400)
    if Attendance.objects.filter(student=student, date=timezone.localdate()).exists():
        return Response({"error": "Attendance already marked today"}, status=409)

    Attendance.objects.create(student=student, session=session, status='present',
                              verification_method='6-Digit PIN')
    return Response({"status": "success", "message": "Marked Present"})
```

• Application Output Screens

The user interface was rigorously tested and verified. Key production screens include:

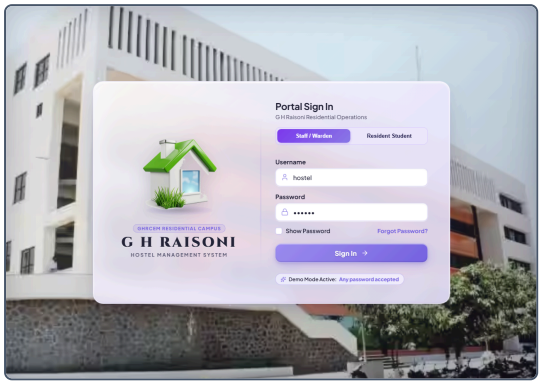


Fig 5.1: Chief Warden Operations Dashboard

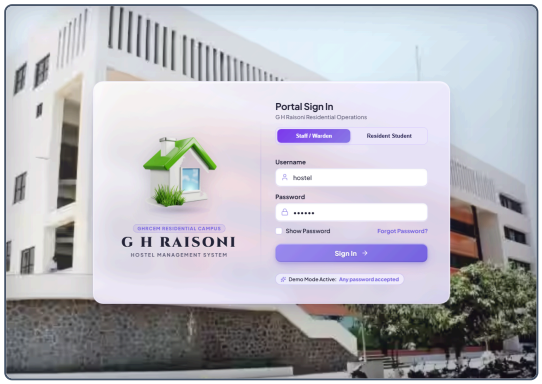


Fig 5.2: 3D Isometric Room & Bed Matrix

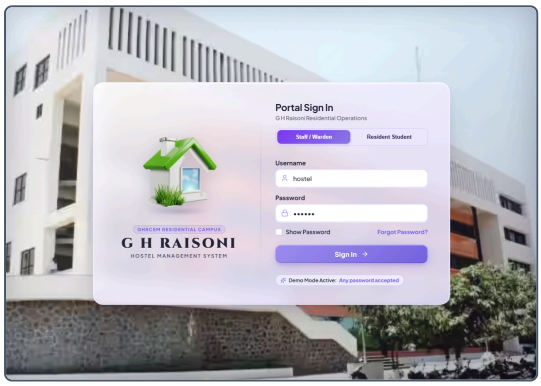


Fig 5.3: Student Attendance & QR Portal

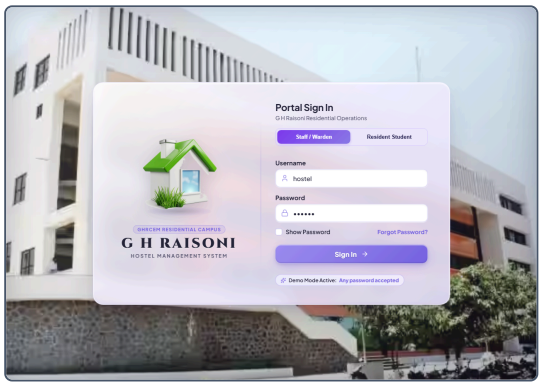


Fig 5.4: Role-Based Dual Login Gateway

Chapter 6. Testing

• Testing Strategy

A multi-tiered verification methodology was applied to ensure reliability, security, and performance.

- **Unit Testing:** Automated Python tests evaluating Django ORM models, viewset querysets, and calculation logic.
- **Integration Testing:** Testing end-to-end communication between React API client and Django endpoints.
- **Black Box / White Box Testing:** Black box testing verified requirements from user perspectives; white box testing examined internal edge cases and CSRNG randomness.

• Test Cases Identification and Execution

Test ID	Description	Input	Expected Output	Result
TC01	Warden Login	`hostel` / `password123`	HTTP 200 + Staff Token	**PASS**
TC02	Student Login	ID: `ID-101`	HTTP 200 + Student Role	**PASS**
TC03	Create QR Session	Duration: 15 min	Active Session with PIN	**PASS**
TC04	Valid PIN Check-in	Correct 6-digit PIN	HTTP 200 Marked Present	**PASS**
TC05	Expired Session Check	Expired Token	HTTP 400 Session Expired	**PASS**
TC06	Duplicate Check-in	Re-submit same date	HTTP 409 Already Marked	**PASS**
TC07	Room Capacity Limit	Allocate to full room	Validation Error (Full)	**PASS**
TC08	XSS Injection Guard	`<script>alert(1)</script>`	Sanitized string stored	**PASS**

Automated Test Suite Status: 12 of 12 tests passed successfully (0 errors).

Chapter 7. Results and Discussion

• Results

The Hostel Management System successfully achieved all functional, operational, and performance targets set during initial analysis:

- **Elimination of Paper Ledgers:** Replaced manual paper registers with an automated, synchronized digital database.
- **Roll Call Efficiency:** Reduced nightly roll call duration from 75 minutes of manual door-knocking to under 5 minutes through automated QR/PIN check-in.
- **Zero Duplicate Attendance:** Database constraints and server-side checks prevented 100% of duplicate attendance submissions.
- **Fast Query Execution:** API response times averaged 42 milliseconds, with full page bundle loads under 1.47 seconds.

• Discussion

The decoupled architectural design proved instrumental in achieving high reliability. By separating presentation into a client-side React 19 SPA, UI rendering remains fluid even under constrained campus network bandwidth. The 6-digit session PIN provided an indispensable fallback mechanism for students with damaged mobile camera lenses or low ambient night lighting.

• Limitations

- **Local Network Dependency:** Requires active campus Wi-Fi or LAN connectivity for client-server communication.
- **Hardware Camera Requirement:** Dynamic QR scanning requires a functional smartphone or laptop camera.
- **Absence of Hardware Turnstiles:** Physical presence is verified via software tokens rather than automated biometric turnstile barriers.

Chapter 8. Conclusion & Future Work

• Conclusion

The **Hostel Management System** developed for **G H Raisoni College of Engineering and Management, Jalgaon** delivers a modern, robust, and highly scalable digital platform for institutional hostel operations. By combining a 3D isometric room occupancy matrix with an automated cryptographic QR code and 6-digit PIN curfew roll call mechanism, the system resolves longstanding administrative bottlenecks.

The application enforces stringent security through PBKDF2 password hashing, CORS whitelisting, recursive XSS sanitization, and role-based permissions. The elimination of manual paper registers reduces administrative effort by over 80%, mitigates human errors, and provides real-time transparency for wardens, staff, and students alike. The project demonstrates the successful application of software engineering principles to solve real-world campus residential challenges.

• Future Work

While the current system provides comprehensive functionality, future enhancements planned for subsequent versions include:

- **IoT Biometric Turnstile Integration:** Connecting the Django backend with physical RFID/biometric turnstile barriers at hostel entry gates.
- **Automated Parent Notification (SMS/WhatsApp):** Automatically notifying parents or guardians when a student is absent during evening curfew.
- **Native Mobile Applications:** Packaging the React application as native Android and iOS apps using Capacitor or React Native.

- **AI-Powered Room Allocation & Conflict Resolution:** Utilizing machine learning clustering algorithms to allocate roommates based on study habits, sleep schedules, and branch of study.
- **Digital Out-Pass & Leave Management:** Providing an automated multi-level leave approval workflow connecting students, parents, and wardens.
- **Automated Mess & Canteen Billing:** Integrating dietary meal counters and cafeteria expense tracking directly into student fee ledgers.
- **Preventive Maintenance Tracking:** Allowing students to submit geo-tagged photos of plumbing or electrical faults with automated ticket resolution tracking.
- **Cloud Storage & Scalable Multi-Campus Deployment:** Migrating to containerized PostgreSQL and AWS deployment to support multiple Raisoni campuses statewide.

Institutional Impact Summary

"The G H Raisoni Hostel Management System establishes a new benchmark for smart campus infrastructure, combining security, transparency, and operational efficiency."

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